

User-Centred design

- Personas
- User profiles
- Scenario / story narrative

When you use filters, you have to check properly the information that you show

The previous experiences are so important when a new system is ~~tested~~

Foundation of user interface design

Human interface - Laws

1. A computer shall not harm your work or, through inaction, allow your work to come to harm
2. A computer shall not waste your time or require you to do more work than is necessary
3. An interface is humane if it is responsive to human needs
4. The user should set the pace of interaction

Sampling of design Methodologies

Human-centered design

User-centered design

Usage-centered design

Cognetics and locus of attention

Historical perspective

1960:

- first GUI
- first OO concepts
- first functional programming concepts

1970:

Xerox Palo alto research
Microsoft incorporated
Apple incorporated

1980:

Microsoft started developing windows,
Technical world, at large, aware of GUI's
Apple meets sales

1990:

Windows 3.0
OO becomes dominate programming paradigms
Windows 95
Windows 98

2000:

Windows XP
GUI dev environments begin to mature
iPhone/Android
Windows 7
Nation, at large, aware of touch interfaces

2010:

Ipad / tablet

HCI Constants

Humans have

Memory
Perceptual System
Cognitive System
Motor System

store information
sensing environment
for processing or thinking
carry out response

Limitations

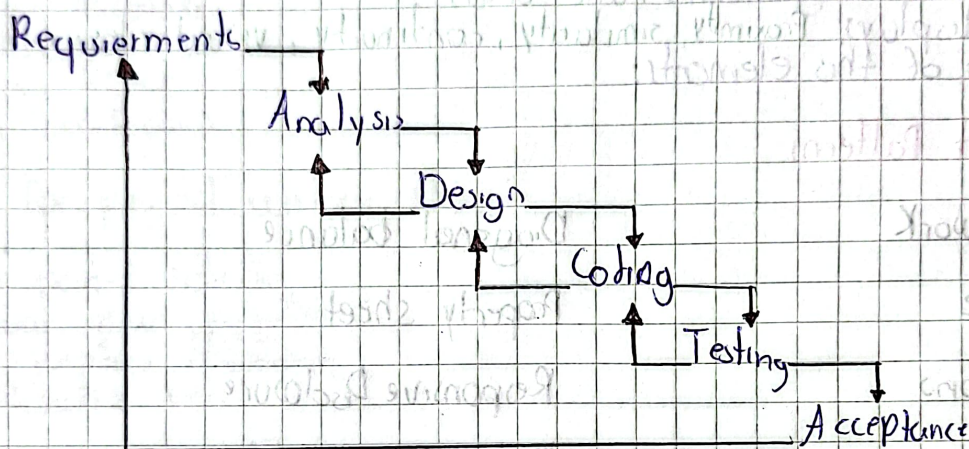
short-term memory
it takes time to retrieve information
humans have a single conscious focus of attention

Design with human factors

Some Key Principles

- Avoid long list of choices
- Avoid modes
- Enable the automation of behaviors that don't require signification
- Easy of learning
- Provide knowledge
- Use terms and concept that familiar to users

SDLC



Getting input from the user

Principles

- Make sure the user understands what's asked and why
- Do usability testing

Control Choice

Factors to consider

- Available space
- Expectations from other apps
- Available technology

Controls

- Checkboxes
- Radio buttons
- Dropdown
- Table
- Single line of text
- Structural format
- Pickers

Patterns

Choice related pattern

- Dropdown chooser
- Illustrated choice
- List builder

Other patterns

- Good defaults
- Same page error messages

Five elements of page layout

- 1- Visual Hierarchy: Define the informational structure
- 2- Visual flow: Sets up focal points
- 3- Grouping and alignment: Make orders
- 4- Dynamic displays: Proximity, similarity, continuity, visual closure
- 5- Integration of the elements:

Page Layout Patterns

Visual Framework

Diagonal balance

Center stage

Property sheet

Titled sections

Responsive Disclosure

Card stack

Responsive enabling

Collapsible panels

Liquid Layout

Movable panels

Grid of equals

Right/left alignment

Accordion

Design Patterns

Common and proven solution to reoccurring problem in some context

Solution → What and how

Problem → When

Context → Why

UI Organizational Patterns

Before the pattern, think about

- Showing one single item
- Showing a list of things
- Providing tools to create
- Facilitating a single task

Patterns for information

- Future, search and browse
- News stream
- Picture manager
- Dashboard
- Settings Editor
- Many workspaces

Patterns for physical window structure

- Two panel selector
- Canvas plus palette
- One-window drilldown
- Alternative view

Other organizational patterns

- Wizard
- Extra on demand
- Intriguing branches
- Multi-level help